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Structural behavior of concrete-filled steel tube short columns stiffened by external and internal continuous spirals

Salih K. Alrebeh[a,b,*], Talha Ekmekyapar[a]

[a] Department of Civil Engineering, Gaziantep University, 27310 Gaziantep, Turkey
[b] Civil Engineering Department, Thi-Qar University, 64001 Thi-Qar, Iraq

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ABSTRACT

Concrete-filled steel tube (CFST) column is one of the most effective forms of composite members in which the steel tube provides both axial strength and confining pressure to improve the compressive capacity and the ductility of the core concrete. This paper presents an evaluation of the structural behavior of CFST short columns strengthened by external and internal continuous spirals. A total of 16 CFST specimens were tested under axial compression. Specimens were stiffened as follows: (1) external continuous spiral (ECS) welded to the exterior surface of the steel tube, (2) internal continuous spiral (ICS) welded to the interior surface of the steel tube, and (3) unwelded internal continuous spiral (UICS) placed longitudinally inside the steel tube. Self-compacting concrete (SCC) was used in the experimental study. The main parameters in this study are the diameter of spiral bar, the number of spiral turns, and the location of the continuous spiral. From the test results, it was concluded that toughness, elastic strength, and ductility were improved significantly as a result of stiffening the CFST columns by external and internal spirals. ECS, ICS, and UICS improved the compression capacity up to 46.8%, 48.7%, and 47.9%, respectively, as compared with the control specimen. A theoretical model to estimate the ultimate load of stiffened CFST columns was developed and close correlations were found between the model and the experimental results.

List of notations

A_c	Concrete net area
A_S	Cross sectional area of steel tube
A_{St}	Cross sectional area of steel material
CFST	Concrete-filled steel tube
D	External diameter of steel tube of CFST column
d	Diameter of spiral
E_C	Concrete modulus of elasticity
ECS	External continuous spiral
ICS	Internal continuous spiral
f'_C	Characteristic 28-day concrete cylinder strength
fcc	Confined concrete strength
fr	Confining pressures
fss	Confining pressure provided by horizontal hoops
fssc	Confining pressure provided by spiral turns
K	Confinement coefficient
K_1	Initial stiffness of CFST column
L	Length of concrete-filled steel tube
LVDT	Linear variable displacement transducer
n	Number of spiral turns
N_1	First peak load
N_2	Smallest measured load after reaching peak
$N_{u,\ exp}$	Experimental ultimate load capacity
N_{pre}.	Predicted ultimate load capacity
P_y	Yield strength of CFST column
S	Spacing of spiral
SCC	Self- compacting concrete
t	Thickness of steel tube
UICS	Unwelded internal continuous spiral
Δ_y	Yield displacement
ζ	Confinement index
ζ_{cr}	Critical Confinement index
σ_{sy}	Yield strength of steel tube
σ_{spy}	Yield strength of spiral
σ_{sz}	Longitudinal stress of steel tube
$\sigma_{s\Theta}$	Hoop stress of steel tube
ε_2	Strain corresponding to N_2
ε_y	Yield strain of test specimens
ε_{co}	Peak strain of unconfined concrete
μ	Strain degradation rate
ψ	Toughness, or energy absorption

* Corresponding author at: Department of Civil Engineering, Gaziantep University, 27310 Gaziantep, Turkey.
E-mail address: sa68988@mail2.gantep.edu.tr (S.K. Alrebeh).

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1. Introduction

CFST short columns are very effective compression members in structures including bridges, buildings, and piled foundations. CFST column combines the favorable properties of steel and concrete. Steel tube usage provides high tensile capacity and improves the ductility, whereas the concrete core increases the compressive strength and stiffness [1–6].

The composite action produces outstanding mechanical properties. The steel tube provides transverse reinforcement to the core concrete, whereas the concrete core delays the local and overall buckling of steel tubes. Moreover, in structures with CFST members, steel tubes could be utilized as formwork and provide shoring during construction, thus reducing construction costs [7–9]. However, almost all researchers agree that imperfect interface bonding between concrete and steel occurs most often along the elastic loading phase. This is the main disadvantage of using CFST columns due to different material properties such as Poisson's ratio, elasticity modulus, etc. [10,11]. As a result, imperfect interface bonding results in a reduction in the axial compressive behavior and elastic stiffness of the CFST column. In order to solve this issue, it is urgent to propose an appropriate form of internal strengthening or external confinement to improve the interfaces bonding during the initial elastic phase.

Different external confinement methods have been developed to strengthen CFST columns, such as FRP jackets [12–14], steel rings [15–17], steel spirals [18] and circular steel jackets [19]. All of the above experimental forms confirmed that using external confinement provides further restricting pressure to the concrete core, thereby improving the uniaxial behavior of the composite column. On the other hand, the concept of using internal stiffeners has been studied by many researchers, such as steel strips [9,20–25], thin-walled liners [26–28], inclined tie bars [29] and tab stiffeners [30]. The confined functionally graded material (FGM) arch is used to stiffen the damaged steel pipes as a liner encased in the host pipe. The buckling behavior of the confined and unconfined FGM arches under external loading was studied analytically and numerically by Li et al. [31]. It was concluded that the pressure capacity of the confined FGM arches is much larger than that of the unconfined arches, indicating the very significant effect of the confinement on the buckling resistance of the FGM arch.

The thin-walled polyhedral polymer pipe liner was proposed by Li et al. [32] for the internal rehabilitation of steel pipes. The elastic buckling behavior of polyhedral polymer liner was studied under uniform external pressure using finite element analysis. The buckling pressure of the polyhedral pipe liners increase with the increment of the thickness-to-radius ratio and the decrement of sides number in polygon base shape. The polyhedral pipe liner can bear uniform external pressure up to 10 times as much as the cylindrical pipe liner. The post-buckling behavior of the polyhedral pipe liner is less stable than that of the cylindrical pipe liner. Li et al. [33] studied the effect of grouting voids on the elastic buckling behavior of confined pipe liners analytically and numerically. Three different void lengths were used in the study. It was concluded that the medium and large grouting voids significantly decrease the critical buckling pressure. The increase of the void thickness decreases the buckling pressure.

Lai and Ho [18] studied the effect of external continuous spirals on the compression capacity of CFST columns. Thirty-eight specimens were tested using different diameters and different spacing. It was drawn that continuous spirals can increase the initial stiffness, strength, ductility, and the interface bonding. Spirals limited the lateral deformation of steel tubes and core concretes and enhanced the imperfect interface bonding.

There is only one research in literature studying the effect of external confinement in the form of ECS to strengthen CFST members, while there is no previous study investigating the effects of internal stiffening using ICS. The aim of this study is to evaluate the structural behavior of CFST columns strengthened by welded ECS, welded ICS, and UICS placed longitudinally inside the specimens. To prove the efficiency of the continuous spiral, 16 CFST short specimens were tested under compression and the results were compared with the control specimen. More attention was paid to study effects of UICS on the mechanical performance of CFST columns. The use of UICS as a stiffening means is an interesting alternative for cost reduction, since the continuous spirals are usually prefabricated, thus saving installation time at the construction site. This research is an important step in the development of researches concerning the enhancement of CFST columns. Finally, an analytical model was proposed to estimate the ultimate axial load of CFST short columns strengthened by the three stiffening schemes.

2. Experimental program

2.1. Details of test specimens

In order to evaluate the uniaxial behavior of CFST columns, sixteen test specimens were manufactured and tested under axial loading. The CFST specimens were divided into five groups according to the location of the continuous spiral (internal or external), diameter of the spiral bar (8, 5.6 mm), number of spiral turns (7, 5, 3), and the installation method (welded or unwelded). Each group has three specimens. The test specimens of the first and second groups were internally stiffened by continuous spirals having 3, 5 or 7 turns with diameter of 8 and 5.6 mm, respectively. Similarly, specimens of third and fourth groups were externally confined by continuous spiral having 3, 5 or 7 turns with diameter of 8 and 5.6 mm, respectively. The test specimens in the fifth group were internally stiffened by unwelded continuous spiral having 3, 5 or 7 turns of 8 mm diameter. For comparison purposes, a CFST short column without external and internal continuous spiral was used as control specimen. Specimen specifications and strengthening details are illustrated in Fig. 1 and Table 1.

A naming system for distinguishing CFST specimens has been established. It consists of four parts: specimen type, spiral location, diameter of spiral (8, 5.6), and number of spiral turns (7, 5, 3). Letter C denotes the CFST specimen and symbol IS denotes the internal spiral while symbol ES denotes the external spiral. For example, (C-ES-8-3) represents a CFST specimen confined by external spiral (denoted by the second part “ES”) with diameter of 8 mm (denoted by the third part “8”) and three turns (denoted by the fourth part “3”) and. The suffix “W” denotes the internal spiral which is not welded to the inner surface of the steel tube. The control CFST column was represented by C—S0 (without external and internal spiral).

2.2. Material properties

All test specimens have the same dimensions with outside diameter of 114.3 mm, total length of 320 mm, steel tube thickness of 3.05 mm, and length to diameter ratio (L/D) of 2.8. The two ends of each tube were finely cut to the desired length using a lathe so that the both ends of the tube were parallel to each other and perpendicular to the specimen longitudinal direction; this is a significant step in the composite column testing process to obtain robust and reliable results [34]. Three coupons were taken longitudinally from steel tubes and tested under tension according to the American Standards and Testing Materials [35] limitations. The average yielding stress (σ_{sy}) of steel tubes is 380 MPa.

All external and internal spirals are made of deformed rebar with two different diameters (8, 5.6 mm). The yield strength of the 8 mm and 5.6 mm bars are 530 MPa and 549 MPa, respectively. The external and internal continuous spirals were welded to the steel tube intermittently using weld stitches spaced at central angle of 90°, while the internal continuous spiral was placed longitudinally inside the test specimens without welding. As reported by [18], each end of the test specimen was equipped with a horizontal hoop having a length at least 1.5 times the circumference of the specimen. Before filling the concrete, the

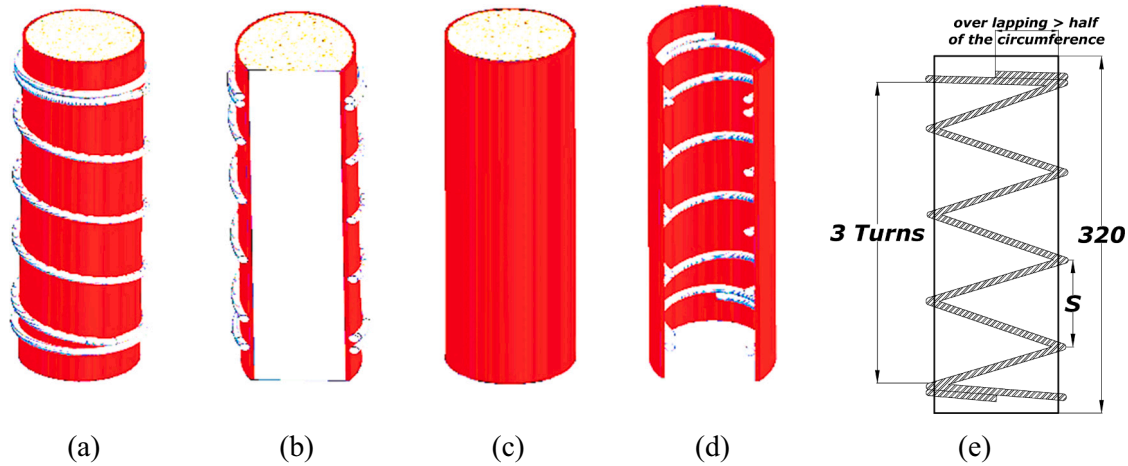


Fig. 1. Typical CFST column stiffened by continuous spiral, a) CFST with external continuous spiral, b) Sectional view of externally stiffened CFST, c) CFST with internal continuous spiral, d) Sectional view of internally stiffened CFST, and e) Typical C-ES-8-3 specimen.

Table 1
Properties of CFST specimens.

Specimen	D × t × L (mm)	Dia. of spirals (mm)	No. of spiral turns	Spacing of Spirals (mm)	$N_{u,\ exp}$ (kN)
C-IS-8-7-W	114.3 × 3.05 × 320	8	7	36	1387.63
C-IS-8-5-W	(Group 5)		5	50	1220.56
C-IS-8-3-W			3	84	1128.69
C-IS-8-7	114.3 × 3.05 × 320	8	7	36	1400.03
C-IS-8-5	(Group 1)		5	50	1294.40
C-IS-8-3			3	84	1098.21
C-IS-5.6-7	114.3 × 3.05 × 320	5.6	7	37	1198.04
C-IS-5.6-5	(Group 2)		5	52	1103.67
C-IS-5.6-3			3	87	1043.12
C-ES-8-7	114.3 × 3.05 × 320	8	7	36	1361.44
C-ES-8-5	(Group 3)		5	50	1312.54
C-ES-8-3			3	84	1090.90
C-ES-5.6-7	114.3 × 3.05 × 320	5.6	7	37	1160.94
C-ES-5.6-5	(Group 4)		5	52	1069.72
C-ES-5.6-3			3	87	1024.68
C-S0	114.3 × 3.05 × 320	–	–	–	960.87

interior and exterior surfaces of the experimental specimen were precisely cleaned from dust and rust residues to ensure appropriate interaction between concrete and steel tube. After that, the glass plate was fastened to the bottom end of the test specimens using silicone to prevent leakage of the cement mortar and to get smooth concrete surface.

SCC containing fly ash was used in the experimental study. SCC terminates the need of compaction process and increase the interaction between concrete and steel tube [3,36]. Moreover, fly ash is a very useful material, which has a spherical micro-structure and provides high flowability. River sand and coarse aggregates (maximum size of 10 mm) were utilized in the mix. Additional materials were also used to produce SCC with the necessary fresh properties including superplasticizers and fine crushed stone. The mixing ratio of SCC is illustrated in Table 2.

Two SCC mixes were prepared and standard cylinders (100 × 200 mm) were used. All molds were filled with three layers of SCC. After completion of concrete casting, all test specimens were cured at laboratory temperature for 28 days. Before performing the test process, the compressive strength test was carried out according to [37] limitations. Four cylinders from each batch were tested. The average compressive strength measured from the two concrete batches was about 41.0 MPa.

It is necessary to make sure that the compressive load is applied evenly to the entire cross section of the specimen and transmitted simultaneously to the concrete and the steel. Therefore, the surface should be regular and horizontal at the top of the test column. In order to achieve this purpose, the last 5 mm from the top of the specimen remained unfilled during the concrete placing process. When the curing step was complete, the top surface of test specimens was filled with a high strength epoxy. Before testing, the glass plate and silicone residue were removed from the underside of the CFST specimen and carefully cleaned. Fig. 2 shows the specimens prepared for testing.

2.3. Test setup and procedure

To examine the structural performance of the CFST columns, all column specimens were tested under a monotonic axial load using the 3×10^5 kN capacity compression testing machine. The axial load was applied through the lower end of the test specimen. The specimens were accurately placed in the testing machine and all precautions have also been taken to prevent any unintentional deviations during the uniaxial load application. During the axial compression test, the average linear deformation of both sides of the tested column was measured using two linear variable differential transducers (LVDTs) as shown in Fig. 3. LVDTs were also linked to the data acquisition system and the load-shortening readings were continuously recorded by the computer to study and evaluate the behavior of CFST short specimens along the pre-collapse and post-collapse phase. For all CFST specimens, load-control displacement was imposed at a rate of 1 mm/min until these specimens reached critical failure.

Table 2
Mix ratio and concrete properties.

Cement (kg/m^3)	Fly ash (kg/m^3)	Coarse aggregates (kg/m^3)	River sands (kg/m^3)	Crush stone (kg/m^3)	Water (kg/m^3)	Superplasticizer (kg/m^3)	f_c' (average) (MPa)	Density (kg/m^3)
275	285	709.83	650.68	216.89	188.10	2.77	41.0	2338.27

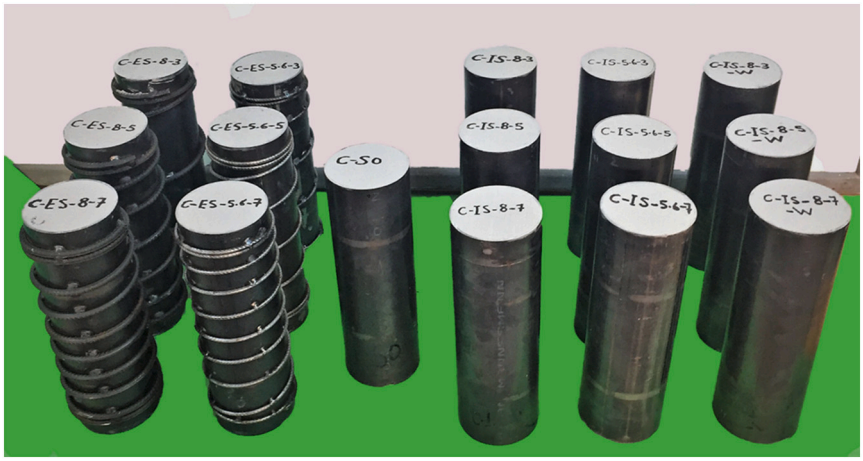


Fig. 2. CFST specimens.

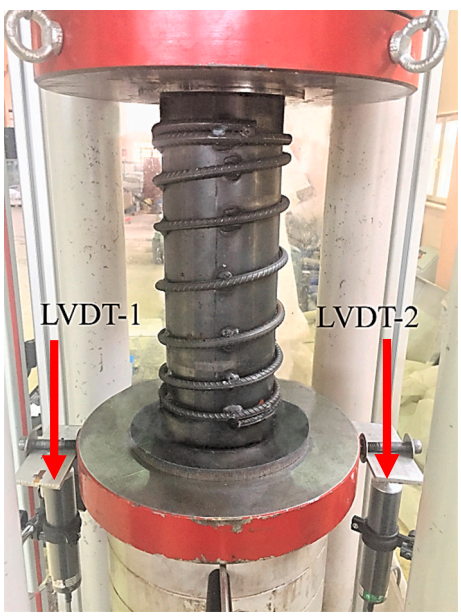


Fig. 3. Column testing.

3. Experimental results and discussion

3.1. Compressive loading–end shortening curves

The measured compressive load is plotted against the deformation for CFST specimens as shown in Figs. 4–9. The axial deformations of capping material are excluded [16]. It can be noted that the compression capacity of spiral-stiffened CFST specimens increases remarkably in the comparison with the control specimen. At first, the axial compressive load increases linearly with the displacement. When the strain increases further, elasto-plastic zone begins that indicates micro-cracks development in the concrete. Steel tube and continuous spiral restrain the expansion of the concrete and thus the confining pressure will be developed. Moreover, since the inward buckling of steel tubes is restrained by the concrete, strain-hardening of steel tubes develops considerably at large strain, which enhances the steel tube strength and hence the confining pressure. It could also be observed in Figs. 4–9 that CFST specimens with large confining pressure (larger diameter and smaller spacing of the continuous spirals) present an ideal strain hardening behavior (stress increases at elasto-plastic stage).

Fig. 4 presents the axial load–deflection response of CFST specimens stiffened by welded ECS. The stiffened specimens C-ES-8-7, C-ES-8-5, C-ES-8-3, C-ES-5.6-7, C-ES-5.6-5, and C-ES-5.6-3 led to performance improvement of 41.6%, 36.5%, 13.4%, 20.8%, 11.3%, and 6.6%, respectively, as compared with the control specimen. Results indicate that compressive capacity and axial stiffness of CFST columns are improved significantly as a result of increasing the number of spiral turns. Moreover, the compressive capacity of columns is developed by increasing the spiral bar diameter.

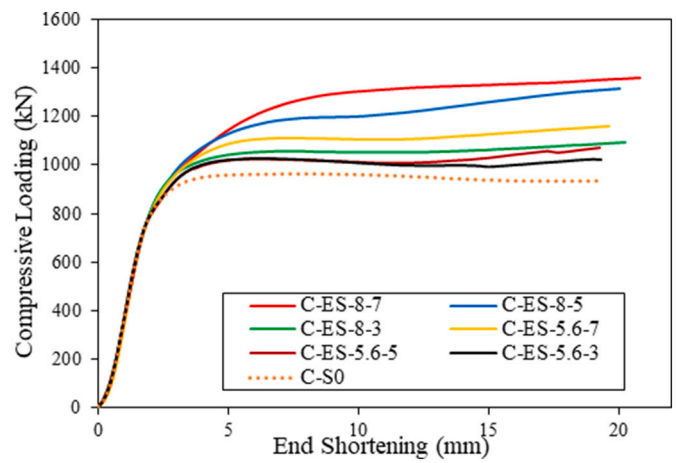


Fig. 4. Load-deformation responses of CFST specimens stiffened by welded ECS (Group 3, 4).

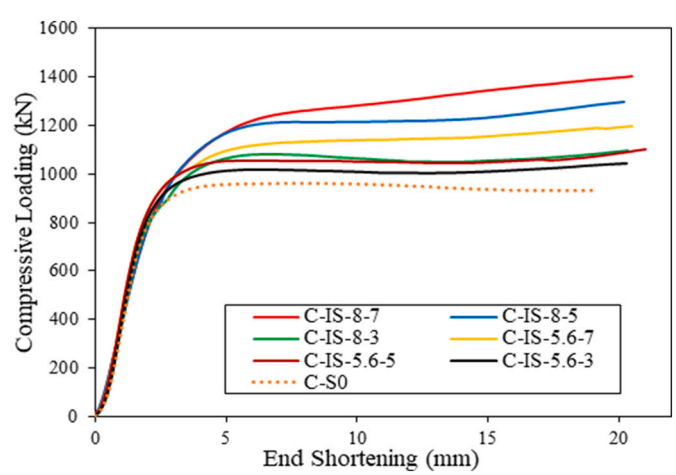


Fig. 5. Load-deformation responses of CFST specimens stiffened by welded ICS (Group 1, 2).

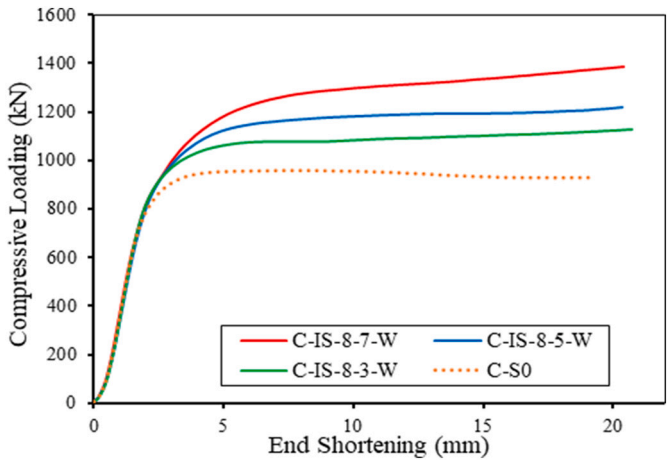


Fig. 6. Load-deformation responses of CFST specimens stiffened by UICS (Group 5).

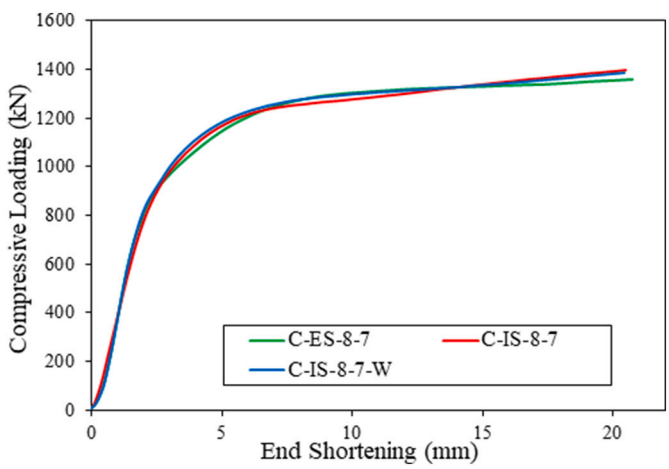


Fig. 7. Load-deformation responses of CFST specimens stiffened by spiral with 7 turns.

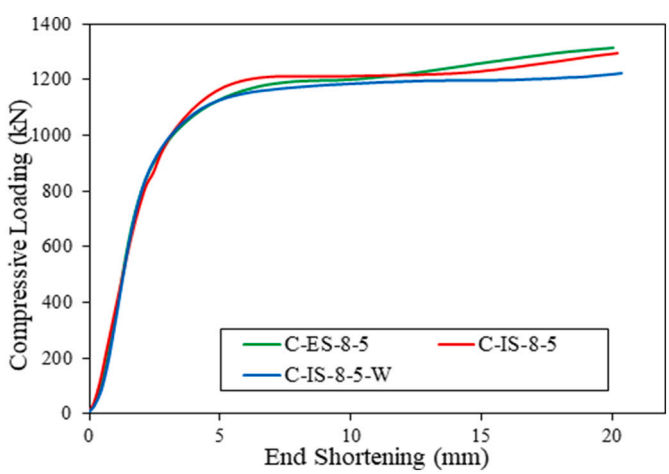


Fig. 8. Load-deformation responses of CFST specimens stiffened by spiral with 5 turns.

The axial load–deflection curve of CFST specimens stiffened by welded ICS is shown in Fig. 5. The stiffened specimens C-IS-8-7, C-IS-8-5, C-IS-8-3, C-IS-5.6-7, C-IS-5.6-5, and C-IS-5.6-3 led to performance improvement of 45.7%, 34.7%, 14.3%, 24.6%, 14.8%, and 8.5%, respectively, as compared with the control specimen. As shown in Fig. 5, the axial compressive strength and axial stiffness of the CFST specimens stiffened by welded ICS increase as the spiral diameter increases and spiral spacing decreases.

Fig. 6 presents the axial load–deflection response of CFST specimens

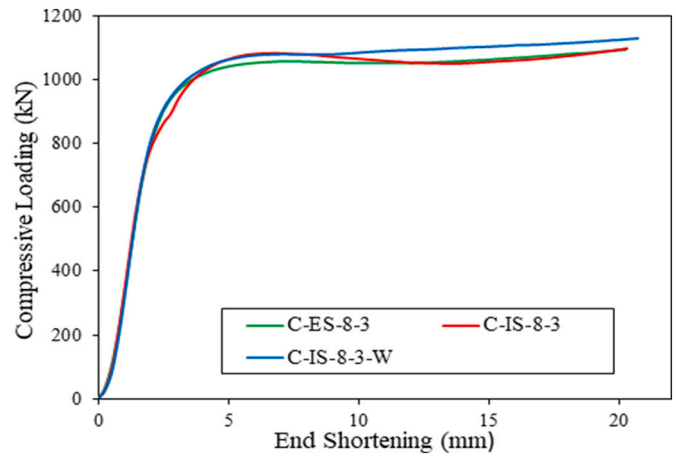


Fig. 9. Load-deformation responses of CFST specimens stiffened by spiral with 3 turns.

stiffened by UICS. The stiffened specimens C-IS-8-7-W, C-IS-8-5-W, and C-IS-8-3-W led to performance improvement of 44.4%, 27.0%, and 17.4%, respectively, as compared with the control specimen. Based on the experimental results, it can be drawn that compressive capacity and axial stiffness of CFST columns are improved significantly as a result of increasing the number of spiral turns.

To compare the effect of UICS, ICS, and ECS on the axial performance of CFST columns, the relation between the axial load and the displacement were plotted. Figs. 7–9 show the relation between the compressive load versus the axial deformation of UICS specimens (C-IS-8-7-W, C-IS-8-5-W, C-IS-8-3-W), ICS specimens (C-IS-8-7, C-IS-8-5, C-IS-8-3), and ECS specimens (C-ES-8-7, C-ES-8-5, C-ES-8-3). As shown in the figures, the compressive capacity of UICS specimens is very close to the capacity of ICS specimens and ECS specimens. For example, the compressive capacity of the UICS specimen C-IS-8-7-W is 1387.63 kN while the compressive capacity of the specimens strengthened by ICS (C-IS-8-7) and ECS (C-ES-8-7) is 1400.04 kN and 1361.44 kN, respectively. The use of UICS as a stiffening means is an interesting alternative for cost reduction, since the continuous spirals are usually prefabricated, thus saving installation and welding time during the fabrication.

3.2. Strain degradation rate reduction

The strain degradation rate represents the ductility of CFST columns to some extent. If the degradation rate of a column is high, the strength decreases speedily with strain, and the column may fail in a brittle manner. If the degradation rate is small, the column exhibits a ductile behavior. To study effects of external and internal spirals on the strain degradation rate (μ) and ductility improvement, μ is calculated using the following formula [16,38]:

$$\mu = -(N_1 - N_2)/(\varepsilon_1 - \varepsilon_2) \quad (1)$$

where $N_1$is the yielding load [16,17,38]. N_1 was calculated using the 1/3 tangent method developed by Park and Choi [39]. N_2 is the load-carrying capacity [16,38] and it is defined as the measured strength at 0.05 axial strain [40]. If there is an ultimate load capacity then N_1 is the first peak load and N_2 is the smallest measured load after reaching peak (slope equals zero); ε_1 and ε_2 are strain corresponding to N_1 and N_2.

The yield strain (ε_y) is calculated using the formula proposed by [41].

Table 3
Strength degradation rate (μ).

Specimens	(ξ)	ε_1	N_1	ε_2	N_2	Strength degradation rate (μ)
C-IS-8-7-W	1.33	0.003648	1174.4	0.05	1346.5	−3.71
C-IS-8-5-W	1.32	0.003645	1078.0	0.05	1196.5	−2.55
C-IS-8-3-W	1.31	0.003641	1017.6	0.05	1107.7	−1.94
C-IS-8-7	1.33	0.003648	1152.0	0.05	1353.5	−4.34
C-IS-8-5	1.32	0.003645	1139.0	0.05	1242.5	−2.23
C-IS-8-3	1.31	0.003641	1013.0	0.05	1067.0	−1.16
C-IS-5.6-7	1.20	0.003598	1067.0	0.05	1164.0	−2.10
C-IS-5.6-5	1.20	0.003598	1003.0	0.05	1060.0	−1.22
C-IS-5.6-3	1.19	0.003597	965.0	0.05	1016.0	−1.10
C-ES-8-7	1.31	0.004013	1160.0	0.05	1336.0	−3.82
C-ES-8-5	1.30	0.004009	1100.0	0.05	1271.0	−3.68
C-ES-8-3	1.29	0.004005	992.0	0.05	1066	−1.60
C-ES-5.6-7	1.19	0.003965	1040.0	0.05	1136.0	−2.06
C-ES-5.6-5	1.19	0.003965	975.0	0.05	1065	−1.94
C-ES-5.6-3	1.18	0.003965	912.0	0.05	998.0	−1.85
C-S0	1.07	0.003918	875.0	0.05	910.0	−0.75

$$\varepsilon_y = \varepsilon_{co} + \left[1200 + 900 * \left(\frac{f'_c}{30} - 1 \right) \right] * \xi^{0.3} \tag{2}$$

$$\varepsilon_{co} = 4.11 \frac{(f'_c)^{0.75}}{E_c} \tag{3}$$

$$E_c = 22 \left(\frac{f'_c}{10} \right)^{0.3} \tag{4}$$

where ε_{co} is the peak strain of unconfined concrete defined in Eq. (3) [42]; E_c is the concrete modulus of elasticity [43]. (ε_y, N_1) is the coordinate of the yield point.

The measured values of N_1, N_2, ε_1, ε_2 and the calculated strain degradation rate μ are shown in Table 3. It can be drawn from the table that the strength degradation rate decreases and the ductility improve by stiffening CFST specimens in the form of external and internal spirals. Moreover, the strength degradation rate decreases as number of spiral turns increases. This is because more turns increase the magnitude and uniformity of confining pressure. Also, the strength degradation rate decreases as diameter of spiral bar increases.

Finally, it is worth noting that all curves shown in Figs. 4–6 tend to ascend after post-elastic region which mean that the external and internal continuous spirals can reduce the strain degradation rate and improve the ductility of the CFST specimens.

3.3. Critical confinement index (ξcr)

Confinement index (ξ) is an important parameter in the composite action as mentioned by [44,45]. Confinement index indicates the ratio of the steel to the concrete strength. In this research, the definition of ξ is widened to consider effects of continuous spiral confinement.

$$\xi = \frac{A_{st}\sigma_{sy}}{A_c f'_c} \tag{5}$$

For external continuous spiral:

$$A_{st} = A_s + \frac{\pi d^2}{4H} \left(2\pi \left(D + \frac{d}{2} \right) + n \sqrt{S^2 + \left(\pi^2 \left(D + \frac{d}{2} \right) \right)} \right) \frac{\sigma_{spy}}{\sigma_{sy}} \tag{6}$$

For internal continuous spiral:

$$A_{st} = A_s + \frac{\pi d^2}{4H} \left(2\pi \left(D - \frac{d}{2} \right) + n \sqrt{S^2 + \left(\pi^2 \left(D - \frac{d}{2} \right) \right)} \right) \frac{\sigma_{spy}}{\sigma_{sy}} \tag{7}$$

where A_{st} is the total cross-section area of steel including the spiral as part of the steel tube thickness [16]; A_s is the steel tube section area; A_c is the concrete net area; σ_{sy}, σ_{spy} and f_c' are the steel tube yielding stress, continuous spiral yielding stress, concrete cylinder strength; d is the diameter of spiral bar; D is the diameter of steel tube; S is the spiral spacing; *n* is the number of spiral turns. Confinement index values of all CFST specimens are listed in Table 3.

The confinement index (ξ) affects the shape of compressive loading–end shortening curves importantly:

(1) For CFST specimens stiffened by UICS: values of ξ are 1.33, 1.32 and 1.31 for specimens C-IS-8-7-W, C-IS-8-5-W, and C-IS-8-3-W, respectively. There are typical yielding points and slope of post-elastic zone increases (i.e. excellent strain hardening behavior).
(2) For CFST specimens stiffened by ICS with diameter 8-mm, the values of ξ are 1.33, 1.32 and 1.31 for specimens C-IS-8-7, C-IS-8-5 and C-IS-8-3, respectively. There are typical yielding points and slope of post-elastic zone increases (i.e. excellent strain hardening behavior).
(3) For CFST specimens confined by ECS with diameter 8-mm, values of ξ are 1.31, 1.30 and 1.29 for specimens C-ES-8-7, C-ES-8-5 and C-ES-8-3, respectively. There are typical yielding points and slope of post-elastic zone increases (i.e. excellent strain hardening behavior).
(4) For CFST specimens stiffened by ICS with diameter 5.6-mm, values of ξ are 1.20, 1.20 and 1.19 for specimens C-IS-5.6-7, C-IS-5.6-5 and C-IS-5.6-3, respectively. There are typical yielding points and slope of post-elastic region increases slightly (i.e. average strain hardening behavior).
(5) For CFST specimens stiffened by ICS with diameter 5.6-mm, the values of ξ are 1.19, 1.19 and 1.18 for specimens C-ES-5.6-7, C-ES-5.6-5 and C-ES-5.6-3, respectively. There are typical yielding points and slope of post-elastic region increases slightly (i.e. average strain hardening behavior).

Critical confinement (ζcr) is the confinement value where the behavior of the CFST column changes from strain softening behavior to strain hardening behavior. The value of (ζcr) equals 1.12 according to [41]. [16] considered that the concrete grade has a significant influence on the (ζcr) and they developed a set of empirical equations correlating (ζcr) value and concrete grade ($0 < f_c' \leq 120$ MPa) as follows:

$$\xi_{cr} = \left\{ \begin{array}{c|c} 0.61 & f'_c \in (0 - 25) \\ 0.0116 f'_c + 0.32 & f'_c \in (25 - 75) \\ 1.19 & f'_c \in (75 - 120) \end{array} \right\} \tag{8}$$

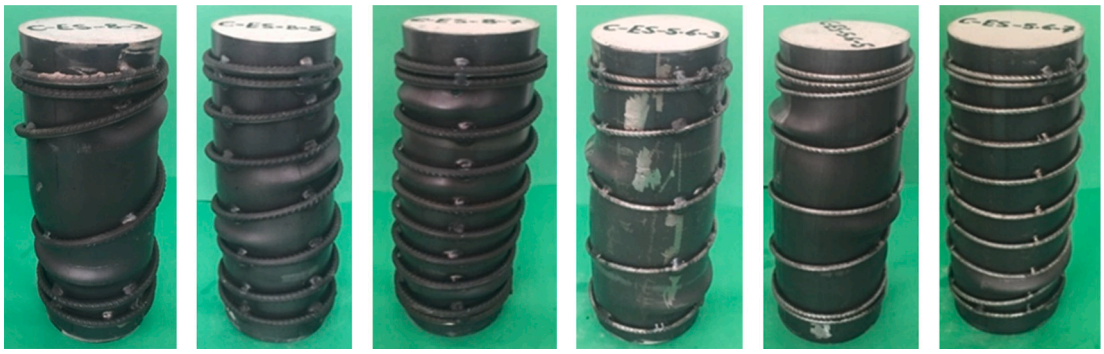
Based on the equation of [16], the ζcr equals 0.81 for the CFST specimens. The value of (ξ) for all stiffened CFST specimens are more than (ζcr) which indicates that the stiffened specimens have a strain hardening behavior.

3.4. Toughness (energy absorption) (ψ)

Toughness, or energy absorption (ψ), is the material capacity to

Table 4
Toughness (energy absorption) (ψ).

Specimens	Toughness(ψ) (kN.mm)	Specimens	Toughness(ψ) (kN.mm)
C-IS-8-7-W	22,737.72	C-ES-8-7	22,429.03
C-IS-8-5-W	20,829.41	C-ES-8-5	21,364.50
C-IS-8-3-W	19,424.45	C-ES-8-3	18,993.91
C-IS-8-7	22,678.03	C-ES-5.6-7	19,858.80
C-IS-8-5	21,370.27	C-ES-5.6-5	18,377.44
C-IS-8-3	18,803.72	C-ES-5.6-3	18,072.84
C-IS-5.6-7	20,575.93	C-S0	17,155.70
C-IS-5.6-5	19,045.30		
C-IS-5.6-3	18,273.21		



(a) (b) (c) (d) (e) (f)

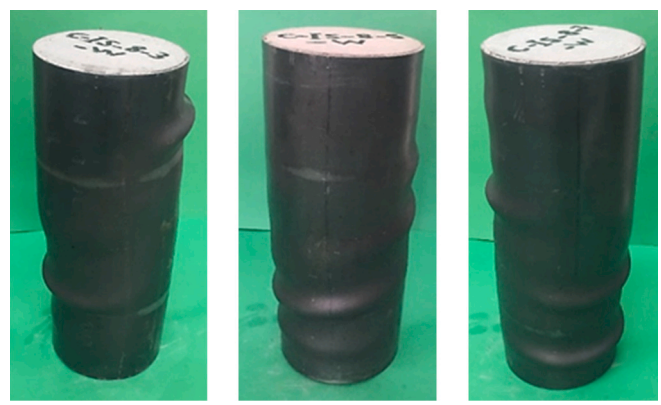
Fig. 10. Failure mode of CFST columns confined by ECS.

absorb energy in the plastic zone before rupturing. Energy absorption is very important for the ductility of structures. Toughness is the entire area under load-deformation curve or stress-strain curve up to a specified value of end shortening; the area can be calculated using numerical calculations or integration [46,47]. Numerical calculations using trapezoidal rule equation are used in this study to calculate the toughness of the CFST specimens up to a specific deformation value. The calculated toughness for all stiffened CFST specimens is listed in Table 4. As shown in the table, the toughness increases in the stiffened CFST specimens in the comparison with the control specimen. Moreover, the toughness increases with the increase of the number of spiral turns and spiral diameter. The specimen C-IS-8-7-W mm yielded the maximum toughness value.

3.5. Failure mode

Failure mode of CFST short columns include yielding of the steel tubes, local buckling of steel tubes, and smashing of the core concrete [48,49]. As shown in Figs. 10 – 12, the specimens strengthened by ECS, ICS and UICS failed due to local outward buckling of the steel tube at the spacing between the turns of ECS, ICS and UICS (not owing to the end effect). This is because ECS, ICS and UICS offer an effective lateral restraint on the steel tube and the concrete, and reduce the effective length of the specimen to the spacing of the continuous spiral. Therefore, the "end effect" owing to the large uniaxial load and the friction strength between the load platen and the steel tube does not dominate this failure.

It is worth noting that the top and bottom surfaces of the core concrete are somewhat higher than that of steel tube (0.5 mm to 1 mm) in CFST specimens stiffened by UICS. It could be interpreted due to stop loading (ending the test) which causes lowering the hoop stresses and therefore pushes inward against the concrete [44]. Fig. 13 illustrates



(a) (b) (c)

Fig. 12. Failure mode of CFST columns stiffened by UICS.

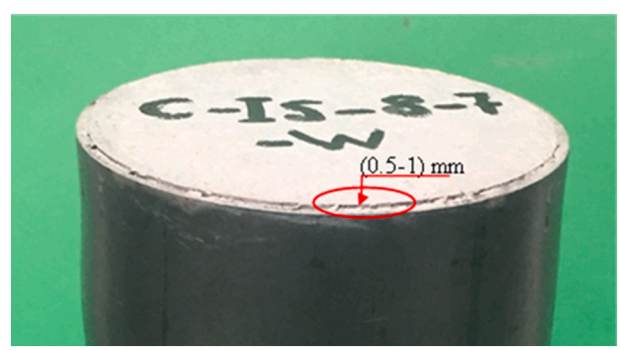


Fig. 13. Slippage between core concrete and tube wall.

(a) (b) (c) (d) (e) (f)

Fig. 11. Failure mode of CFST columns stiffened by ICS.

Table 5
Load capacity improvement rates.

Specimen	Load-carrying capacity (kN)	Improvement ratio (%)	Specimen	Load-carrying capacity (kN)	Improvement ratio (%)
C-IS-8-7-W	1346.5	47.9	C-ES-8-7	1336.0	46.8
C-IS-8-5-W	1196.5	31.4	C-ES-8-5	1271.0	39.6
C-IS-8-3-W	1107.7	21.7	C-ES-8-3	1066.0	17.1
C-IS-8-7	1353.5	48.7	C-ES-5.6-7	1136.0	24.8
C-IS-8-5	1242.5	36.5	C-ES-5.6-5	1065.0	17.0
C-IS-8-3	1067.0	17.2	C-ES-5.6-3	998.0	9.6
C-IS-5.6-7	1164.0	27.9	C-S0	910.0	
C-IS-5.6-5	1060.0	16.4			
C-IS-5.6-3	1016.0	11.6			

this phenomenon in one of the three specimens stiffened by UICS.

3.6. Load-carrying capacity and axial stiffness enhancement

Table 5 summarizes the load-carrying capacities of short CFST columns enhanced by three stiffening schemes. The load-carrying capacity is interpreted as the measured load at 0.05 axial strain [40]. It can be noted from the table that the load-carrying capacity of the experimental columns strengthened by the ICS are higher than those of corresponding columns confined by ECS or strengthened by UICS. ICS, UICS, and ECS improve the load-carrying capacity up to 48.7%, 47.9%, and 46.8%, respectively, as compared with the control specimen.

It can be interpreted from results that ICS and UICS specimens have better load-carrying capacity than ECS specimens and this is due to the fact that ICS and UICS increase the bond between steel tube and core concrete interfaces and increase the buckling resistance of steel tube during the elastic loading phase.

The elastic axial stiffness is a critical parameter for evaluating the structural performance of CFST columns. The axial stiffness (K_1) in this research was calculated using the 1/3 tangent method developed by Park and Choi [39]. (K_1) is defined as follows:

$$K_1 = \frac{P_y}{\Delta_y} \tag{9}$$

where Δ_y represents the yield displacement corresponding to the yield strength P_y.

Table 6 shows the axial stiffness values of the CFST specimens. From the Table, the internal and external continuous spirals can provide early confining pressure along the spiral turns and hence limit the lateral dilation of the steel tube and the core concrete to some extent. Also, it can be observed that the provision of continuous spirals increases the elastic stiffness up to 7.8%, compared to the control specimen.

Table 6
Axial stiffness improvement rates.

Specimen	Axial stiffness (kN/mm)	Improvement rate (%)	Specimen	Axial stiffness (kN/mm)	Improvement rate (%)
C-IS-8-7-W	491.5	7.85	C-ES-8-7	485.3	6.49
C-IS-8-5-W	455.0	1.00	C-ES-8-5	463.8	1.77
C-IS-8-3-W	465.6	2.17	C-ES-8-3	452.0	1.00
C-IS-8-7	471.0	3.35	C-ES-5.6-7	471.8	3.53
C-IS-8-5	460.4	1.03	C-ES-5.6-5	460.0	1.00
C-IS-8-3	452.1	1.00	C-ES-5.6-4	456.4	1.00
C-IS-5.6-7	475.0	4.23	C-S0	455.7	1.00
C-IS-5.6-5	459.1	1.00			
C-IS-5.6-3	462.0	1.38			

4. Proposed model and verifications

A theoretical model proposed by [18] for estimating the ultimate axial load of spiral-stiffened CFST members is adjusted to consider the confining effects given by steel tube, external and internal spirals, and horizontal hoops.

4.1. Proposed model

Von-Mises failure criterion (Eq. (10)) can be utilized to relate the longitudinal stress (σ_{sz}) and the hoop stress ($\sigma_{s\theta}$) of the steel tube at failure to the uniaxial yielding strength. The radial stress of steel tube could be ignored [15,16,18,50].

$$\sigma_{s\theta}^2 + \sigma_{s\theta}\sigma_{sz} + \sigma_{sz}^2 = \sigma_{sy}^2 \tag{10}$$

where σ_{sy} is the uniaxial yielding stress of steel tube.

For the core concrete, the following formula is developed to calculate the confined concrete stress under confining pressure (f_r):

$$f_{cc} = f'_c + Kf_r = f'_c + 3.16\, f_r \tag{11}$$

where f_{cc} is the confined concrete strength, fc' is the cylinder strength of unconfined concrete and (K) is the confinement coefficient, and (K) equals 3.16 as suggested by [51].

The confining pressures could be expressed by drawing the free body diagram as shown in Figs. 14–15. Because the spiral contains spiral turns and end horizontal hoops, the confining effects of the continuous spiral are divided into end part and middle part over the total height of CFST columns.

$$f_r(D-2t)H = 2\sigma_{s\theta}tH + 4f_{ss}\frac{\pi}{4}d^2 + 2\frac{H}{s}f_{ssc}\frac{\pi}{4}d^2 \tag{12}$$

where f_{ss} is the confining pressure provided by horizontal hoops and $f_{ssc}$is the confining pressure provided by spiral turns (i.e. component of f_{ss}).

If the CFST short column reaches the ultimate load, the above equation can be converted to:

For CFST columns confined by external continuous spiral:

$$f_r = \sigma_{s\theta}\frac{2t}{D-2t} + \sigma_{spy}\frac{\pi d^2}{H(D-2t)} + \sigma_{spy}\cos\left[\arctan\left(\frac{S}{\pi\left(D+\frac{d}{2}\right)}\right)\right]\frac{\pi d^2}{2S(D-2t)} \tag{13}$$

For CFST columns stiffened by internal continuous spiral:

$$f_r = \sigma_{s\theta}\frac{2t}{D-2t} + \sigma_{spy}\frac{\pi d^2}{H(D-2t)} + \sigma_{spy}\cos\left[\arctan\left(\frac{S}{\pi\left(D-\frac{d}{2}\right)}\right)\right]\frac{\pi d^2}{2S(D-2t)} \tag{14}$$

where $\sigma_{spy}$is the yielding stress of external and internal spirals, which is equal to 530 MPa and 549 MPa for 8 mm and 5.6 mm diameter, respectively.

The empirical equation developed by [16] is employed herein to evaluate the hoop stress of steel tube:

$$\frac{\sigma_{s\theta}}{\sigma_{sy}} = \begin{cases} 0 & 0 \le \xi < 0.03 \\ -0.003\xi^2 + 0.175\xi - 0.006 & 0.03 \le \xi < 2.5 \\ 0.244 & \xi \ge 2.5 \end{cases} \tag{15}$$

where ξ is the confinement index as defined in Eq. (5).

Lastly, the ultimate load capacity $N_{pre.}$ of CFST short columns strengthened by the external and internal continuous spiral is:

$$N_{pre.} = f_{cc}A_c + \sigma_{sz}A_s + 2n\sigma_{spy}\frac{\pi}{4}d^2\sin\left[\arctan\left(\frac{S}{\pi\left(D\pm\frac{d}{2}\right)}\right)\right] \tag{16}$$

Eq. (16) offers design criteria for the CFST columns stiffened by external and internal continuous spirals which can be expressed as a

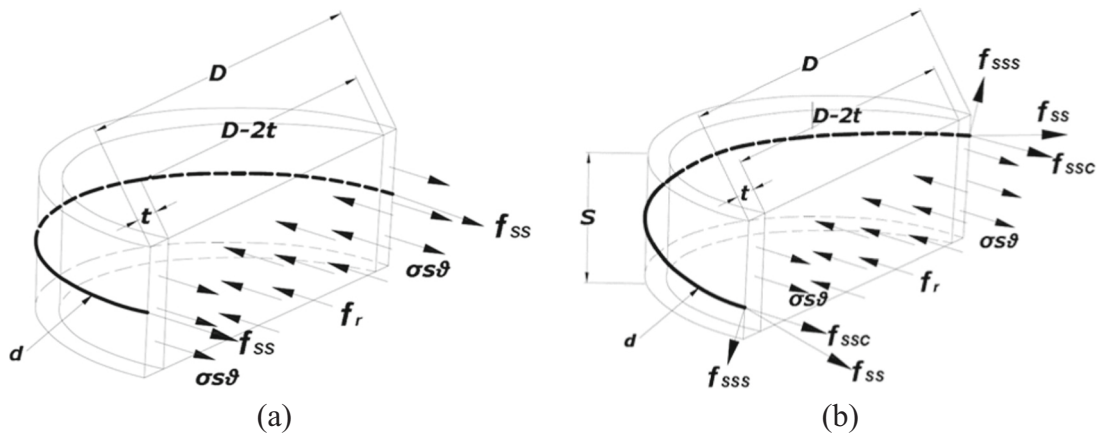


Fig. 14. Free body diagram of CFST short column confined by ECS, a) Details of CFST column with horizontal hoop, b) Details of CFST column with spiral.

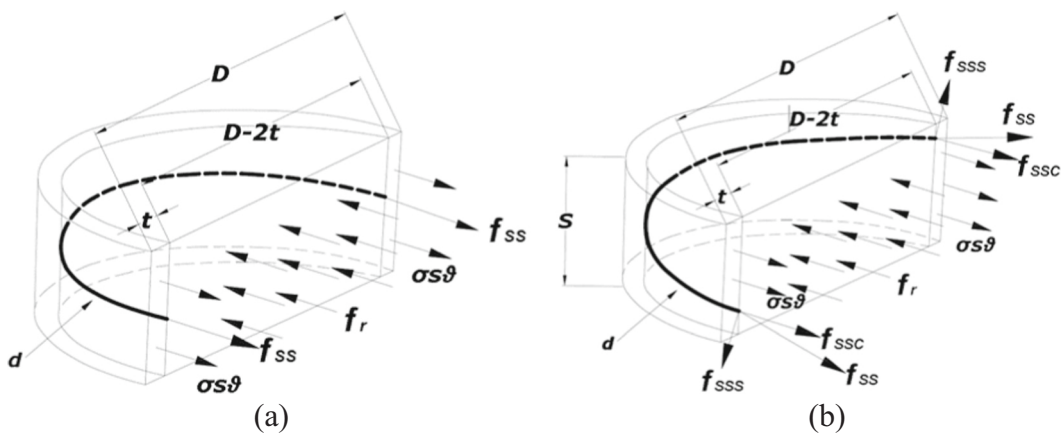


Fig. 15. Free body diagram of CFST short column stiffened by ICS and UICS, a) Details of CFST column with horizontal hoop, b) Details of CFST column with spiral.

Table 7
Experimental and estimated ultimate load of spiral-stiffened CFST columns.

Specimens	$N_{u,\ exp.}$ (kN)	$N_{pre.}$ (kN)	$N_{u,\ exp.}/N_{pre.}$
C-IS-8-7-W	1387.63	1367.93	1.01
C-IS-8-5-W	1220.56	1256.87	0.97
C-IS-8-3-W	1128.69	1141.77	0.97
C-IS-8-7	1400.03	1367.93	1.02
C-IS-8-5	1294.40	1256.87	1.03
C-IS-8-3	1098.21	1141.77	0.96
C-IS-5.6-7	1198.04	1111.3	1.07
C-IS-5.6-5	1103.67	1054.98	1.04
C-IS-5.6-3	1043.12	1003.41	1.04
C-ES-8-7	1361.44	1387.65	0.98
C-ES-8-5	1312.54	1274.64	1.03
C-ES-8-3	1090.90	1156.50	0.95
C-ES-5.6-7	1160.94	1119.24	1.03
C-ES-5.6-5	1069.72	1061.90	1.00
C-ES-5.6-3	1024.68	1009.19	1.01
Maximum value			1.07
Minimum value			0.95
Average value			1.00
Standard deviation			0.0347

function of the confining pressure, materials' properties and other geometric parameters.

In practical design of the CFST column stiffened by external and internal continuous spirals, the ultimate load capacity can be determined by the following procedure:

1. Determine the (ξ) by Eq. (5) and hence the hoop stress $\sigma_{s\theta}$ in the steel tube by Eq. (15).
2. Determine the axial stress σ_{sz} in steel tube by Eq. (10)
3. Determine the confining pressure f_r by Eq. (13) or (14) with the evaluated $\sigma_{s\theta}$ in Eq. (15).
4. Determine the enhanced confined concrete stress f_{cc} using Eq. (11).
5. Lastly, the ultimate load capacity of the CFST column is given by Eq. (16).

4.2. Comparisons and summaries

To evaluate the accuracy of the proposed equations on estimating the ultimate load capacity of stiffened CFST columns, the calculated ultimate loads were compared to the ultimate experimental loads $N_{u,\ exp}$ as shown in Table 7. It can be concluded that the proposed model predicted accurately the ultimate load capacity of CFST specimens stiffened by external and internal continuous spirals. The ratio of estimated ultimate load to experimental ultimate load varies from 0.95 to 1.07, and the standard deviation equals 0.0347. Therefore, the proposed formula is applicable for estimating the ultimate load capacity of spiral-stiffened CFST columns.

For the specimens stiffened by ICS with spiral diameter of 8 mm, the ratio between experimental results and its predictions ranged between 0.96 and 1.03, with an overall average of 1.00 and with a standard deviation of 0.037, while for the specimens stiffened by ICS with spiral diameter of 5.6 mm, the ratio between experimental results and its predictions ranged between 1.04 and 1.07, with an overall average of 1.05 and with a standard deviation of 0.017. These results are conservative predictions, and the ultimate load capacities are acceptable for CFST specimens stiffened by ICS, with an average difference of 5% on the safe side. For the specimens stiffened by UICS, the ratio between experimental results and its predictions

ranged between 0.97 and 1.01, with an overall average of 0.98 and with a standard deviation of 0.023, the proposed method predictions are un-conservative by 0.2% on average. In addition, the ratio between experimental results and its predictions ranged between 0.95 and 1.03, with an overall average of 0.98 and with a standard deviation of 0.040 for the specimens stiffened by ECS with spiral diameter of 8 mm, the proposed method predictions are un-conservative by 0.2% on average, while for the specimens stiffened by ECS with spiral diameter of 5.6 mm, the experimental results and its predictions ratio ranged between 1.0 and 1.03, with an overall average of 1.01 and with a standard deviation of 0.015. These results are conservative predictions with an average difference of 1% on the safe side. Furthermore, it is applicable for estimating the ultimate load capacity of spiral-stiffened CFST columns.

5. Conclusions

This paper presents an evaluation of the structural behavior of CFST short columns strengthened by external and internal continuous spirals. Based on the experimental results, the following conclusions can be drawn:

1- Based on findings, ECS, ICS, and UICS improve the compression capacity up to 41.6%, 45.7%, and 44.4%, respectively, as compared with the control specimen.
2- The compressive capacity of UICS specimens is very close to the capacity of ICS and ECS specimens. The use of UICS as a stiffening means is an interesting alternative for cost reduction, since the continuous spirals are usually prefabricated, thus saving installation and welding time at the construction site.
3- Results indicate that the compressive capacity of CFST columns is improved significantly as a result of increasing the number of spiral turns and spiral bar diameter.
4- From the test results, it could be drawn that strain degradation rate decreases and the ductility improves by stiffening CFST specimens using external and internal spirals.
5- The values of (ξ) for all stiffened CFST specimens are more than (ζcr) which indicates that the stiffened specimens have a strain hardening behavior.
6- Toughness increases in the stiffened CFST specimens compared to the control specimen. Moreover, the toughness increases with the increase of the number of spiral turns and spiral bar diameter.
7- CFST specimens strengthened by ECS, ICS and UICS have failed due to local outward buckling of the steel tube at the spacing between the spiral turns.
8- Based on the findings, ICS, UICS, and ECS improve the load-carrying capacity up to 48.7%, 47.9%, and 46.8%, respectively, as compared with the control specimen. It can be interpreted from results that ICS and UICS specimens have better load-carrying capacity than ECS specimens and this is due to the fact that ICS and UICS increase the bond between steel tube and core concrete interfaces and increase the buckling resistance of steel tube during the elastic loading phase.
9- A theoretical model to predict the ultimate load of stiffened CFST columns was developed and close correlations were found between the model and the experimental results. The ratio of the predicted ultimate load to experimental ultimate load varies from 0.95 to 1.07, and the standard deviation equals 0.0347.

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